

Pavel Petráček

PhD in Autonomous Flying Robotics
Robotics engineer & developer

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Link to academic CV
Google Scholar

About me

Robotics R&D engineer and developer with 10 years of hands-on experience in autonomous UAV systems, multi-agent systems, and real-world robotics. Experienced in taking research from concept to deployment in high-impact environments of subterranean S&R and heritage preservation. Technical expert with strong academic background and project leadership.

Fast learner open to new things.

Core skills

GNSS-denied autonomy
Perception, SLAM, 3D mapping
Sensor fusion (LiDAR, visual, IMU)
Multi-agent systems
Technical and R&D leadership
Systems engineering

- C++, Python, ROS, PX4, git
- Docker, Singularity, Gazebo
- RTOS/embedded targets

Languages

Czech (native)
English (fluent)

Experience

Independent (self) *1/2026–present*
Robotics software development and consultations

Fly4Future s.r.o. (startup) *4/2023–12/2025*
R&D Projects Lead & Hands-On Developer

- Led robotics R&D team of 6
- Delivered autonomous drones to **Temelín nuclear power plant**
- Delivered PoCs and autonomous products to clients worldwide

Multi-Robot Systems @ CTU (lab) *2016–12/2025*
Researcher & Developer

- Fundamental & applied research
- Technical solutions, R&D PoCs, projects leadership & admin, mentoring, industrial cooperation, public demos, teaching
- Open-source contributor to **MRS UAV System**

Selected projects

DARPA Subterranean Challenge (link)

- Subterranean search & rescue with autonomous robot teams
- Responsible for system design, GNSS-denied autonomy, mapping, SLAM, experimentation, integration, and deployment of UAVs
- Winning $1M while competing with top world institutions

Dronument (video)

- Documenting interiors of historical monuments with multiple co-operating UAVs in a synchronized multi-robot formation
- Achieved reliable operation and deployed the autonomous system in 20+ historical sites (incl. 2 UNESCO sites)

Multi-UAV swarming (video)

- Deployed first fully-decentralized swarms w/o communication

DOFEC (video)

- Designed onboard fire detection & localization with autonomous mission execution and fire extinguishment

Education & Research

PhD, Autonomous Flying Robotics CTU, 2019–2024

- Topic: UAV autonomy in perception-degraded settings (pdf)
- 19 publications; h-index: 12 (WoS), 16 (GScholar)

BSc & MSc, Cybernetics CTU, 2014–2019

Honors & Awards

Werner von Siemens prize (web, video) 2025
#1 dissertation out of top (243) 2023–2024 STEM works in Czechia

Joseph Fourier prize (web in Czech) 2025
#1 dissertation out of top (65) 2024 CS works in Czechia

Best paper award at IEEE RA-M (web, paper) 2025
Robotics & Automation Magazine award for our **Dronument** work

AI Awards (web + our tech in the EU annual report) 2025
Team award for contributions in Czech AI R&D

Dean's prize (link) 2024, 2019, 2017
All my theses (dissertation, master's & bachelor's) were selected in top 1% works at CTU in their respective category and year

Czech National Excellence Award M17+ 2022
Excellent international evaluation of our Dronument solution (link)